A Game Theoretic Decision-Making Framework With Conflict-Aware Nash Equilibrium Selection for Autonomous Vehicles at Uncontrolled Intersections

Yuxiao Cao, Xiangrui Zeng, *Member, IEEE*, and Zhouping Yin

***Abstract*— Uncontrolled intersections, as a typical urban traffic environment, are challenging scenarios for autonomous driving due to the potential conflicts and lack of co-ordination between traffic participants. Even if both vehicles are rational players, the existence of multiple Nash equilibira in this dynamic game may result in no consensus between them during the interaction. This may lead to unsmooth rides with late emergency brakes. This paper proposes a conflict-aware game theoretic decision-making framework for autonomous vehicles at uncontrolled intersections considering the Nash equilibrium selection. Combining dynamic programming and iterative best response, a dynamic closed-loop Nash equilibrium calculating method is proposed. The issue of multiple equilibria is analyzed. An equilibrium selection mechanism is designed based on sequentially-constructed Conflict-Aware Bimatrix Games (CAB-Games) and the idea of risk dominant equilibria. The proposed policy provides a relatively smooth drive without being too aggressive or conservative. It is demonstrated in simulations and driving simulator experiments that the policy can interact well with other autonomous driving policies as well as human drivers.**

***Index Terms*— Autonomous vehicles, uncontrolled intersection, decision-making, game theory, equilibrium selection.**

I. Introduction

A. Background

AUTONOMOUS vehicles (AVs) have many potential advantages, such as reducing transport costs, improving safety, and reducing energy consumption [1]. AVs will share the road with other intelligent individuals. They need to both avoid conflict and safeguard their own interests in the interaction with others.

A challenging scenario for AV is the uncontrolled intersections due to their potential conflicts and lack of consensus between traffic participants [2]. Decision-making at intersections can be divided into non-interaction-aware methods and interaction-aware methods. Non-interaction-aware methods are mostly based on predictive control [3], i.e., predicting the trajectories of other traffic participants and considering them as the constraints of the ego vehicle planning. However, these approaches only take into account the influence of other participants on the ego vehicle unilaterally, without considering the influence of the ego vehicle's behavior on other traffic participants [4]. This absence often leads to overly conservative policies to the extent that AVs frequently yield to others, which also affects traffic efficiency in turn. Most interaction-aware methods are based on the concept of game theory [5], [6]. The target of AV policies in games is usually the Nash equilibrium (NE), from which no participant can gain any expected payoff if they choose to deviate. The interaction between vehicles is a dynamic game, where each agent needs to make sequential decisions. Decision making dynamically in interaction scenarios is a difficult task. Therefore many studies on applying game theory in autonomous driving similarly simplify traffic scenarios and approximations of the NE (such as level-k theory) are also used in practice to design decision-making policies [7], [8].

One important but sometimes overlooked aspect for game theoretic AV decision-making is the existence of multiple NEs and its impact for dynamic games. In theory, each game has at least one equilibrium, and some have many [9]. In the uncontrolled intersection scenarios, multiple NEs may exist when the two vehicles are about to arrive at the intersection at the same time (see examples in Section IV-A). Defining and selecting the best NE in a game for decision-making is generally a difficult task [10]. In dynamic games such as the uncontrolled intersections, multiple NEs further complicate the decision-making as agents need to make multiple decisions sequentially. *The consequence of two agents selecting different NEs is that they both will face a new dynamic game at the next decision-making moment*. It is natural to consider this new game when one makes decisions at the moment with multiple NEs. Yet, to the best of the authors' knowledge, this has rarely been addressed in the field of autonomous driving.

This paper presents a game theoretic decision-making framework at uncontrolled intersections considering conflict-aware NE selection. Based on dynamic programming and iterative best response, the closed-loop NEs are calculated offline for AVs' longitudinal decision-making. The multiple

Received 5 January 2024; revised 28 May 2024, 3 September 2024, and 10 October 2024; accepted 15 October 2024. Date of publication 29 October 2024; date of current version 9 January 2025. This work was supported by the National Natural Science Foundation of China under Grant 52272416 and Grant 52188102. The Associate Editor for this article was C. Lv. *(Corresponding author: Xiangrui Zeng.)*

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Ethics Committee of Huazhong University of Science and Technology Drug Clinical Trials.

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Digital Object Identifier 10.1109/TITS.2024.3483844

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NE scenarios are handled using an NE selection mechanism based on sequentially-constructed Conflict-Aware Bimatrix Games (CAB-Games). Each action pair of the CAB-Game corresponds to a closed-loop NE of the original dynamic game. If the CAB-Game has only one NE, the action of this NE is selected. If the CAB-Game also has multiple NEs, the action of the risk dominant equilibrium is selected. For cases where the two vehicles have the same state and the risk dominant equilibrium does not exist, the action is determined online based on the other vehicle's action history. Combining the NE calculation and NE selection with a safety check module, an implementable policy for autonomous vehicle at uncontrolled intersections is obtained. AVs with the proposed policy can make optimal decisions through feedback during dynamic interactions in real time. From the simulation and driving simulator experiments, it is demonstrated that the proposed method performs well in interaction with human-driven vehicles and different intelligent driving policies such as model predictive control (MPC), level-k, and itself. Meanwhile, it not only reduces the average cost by about 20% compared to the MPC and level-1 policies, but also reduces the emergency braking rate by 40% compared to the level-2 policy.

B. Literature Review

1) AVs at Uncontrolled Intersections: Uncontrolled intersections, as a typical scenario of urban traffic, have always been a challenge for decision-making due to the complexity of the traffic participants and the high dynamics of the scenario. Classical decision-making policies at intersections are primarily based on non-interaction-aware predictive control. Chen et al. proposed an autonomous driving strategy at T-intersections using a finite-state machine (FSM) based motion planner [11]. As an improvement of FSM, Hierarchical Finite State Machine model is used for autonomous driving decision-making and performs well in obstacle avoidance [12]. However, these classical methods require artificially defined rules, which consequently can only be used in simple scenarios.

The application of learning-based methods to autonomous driving has become increasingly popular in recent years. A driver intention prediction model for general intersections is presented in [13], and a comparison of turn prediction results based on hidden Markov models (HMM), support vector machines and dynamic Bayesian networks is carried out. In [14], a continuous HMM is developed to predict uncertain intentions and a partially observable Markov decision process is used to model the general decision-making framework at uncontrolled intersections. Nonetheless, these methods perform poorly without overlaying driving data. On the other hand, Reinforcement learning (RL) is a promising solution for driving policies, predictive perception and path and motion planning [15]. David et al. explored the effectiveness of deep reinforcement learning (DRL) for dealing with intersection problems [16], and a DRL method based on deep convolutional fuzzy system is proposed for processing unprotected left-turn scenarios on urban roads [17]. Although learning-based policies are suitable for complex dynamic scenarios, learned policies have high training costs and are difficult to implement for semantic interpretation [18]. Most importantly, many of them still lack in consideration of the impact of the ego vehicle on the surrounding traffic participants [19].

Game theory is a suitable approach for considering bi-directional interactive behaviors and the post-game policies are interpretable [6], thus many researchers have applied it to the interaction between autonomous vehicles. In [20], a hierarchical and game-theoretic decision-making policy for connected autonomous vehicles is proposed to plan optimal short-term and long-term trajectories via vehicle-to-everything. In order to validate and compare the decision-making and control systems of various autonomous vehicles, Zhang and Hofbauer developed a game-theoretic traffic model [9]. A dynamic game policy based on a leader-follower formulation is proposed in [21] and validated by simulation in multiple intersection scenarios. Hang et al. design a cooperative decision-making framework for connected autonomous vehicles for the vehicle multi-lane merging problem using a coalitional game approach [22]. Another trendy method is the level-k game method [23], hierarchising participants and iteratively updating participant policies to find a NE. In [24], a hierarchical game-theoretic decision-making framework is used to simulate drivers' decisions and interactions in traffic, with vehicles assumed to be at different levels. Most of the existing studies on intersection games are based on either static games or dynamic games combined with predictive control. Many of them simply ignore the possibilities of multiple equilibria and assume that a single NE is found for every case [25]. The existence of multiple NEs is a well-recognized problem in game theory. Yet, it is not a well-addressed problem in dynamic decision-making in autonomous driving, and it has been largely ignored in autonomous driving actually.

2) NE Selection in Games: NE selection is a challenging task and has been at the forefront of game theory in recent years. According to the definition of NE, the policy adopted by any player is the best response under the combination of policies adopted by all the remaining players. However, there is often more than one NE in a game, e.g., the Prisoner's Dilemma [26], the Stag hunt game, the chicken game [27], etc. Chicken Game reflects an important and general type of conflict situation that can arise in the real world and it is a simple model of a confrontation of two players on a collision course. There have been some studies on equilibrium selection. One class of equilibrium selection methods is the inductive approach, which generally uses learning or evolutionary dynamics to predict equilibrium [9]. Haruvy and Stahl propose an inductive selection principle based on simple learning dynamics and demonstrate superior predictive performance [28]. Another class of equilibrium selection methods is the deductive approach, generally assuming that participants have beliefs that are consistent with equilibrium. The risk-dominated equilibrium was proposed by Harsanyi and Selten in 1988 [29], and many other researchers have proposed its improvements, such as Bayesian Nash equilibrium [30] and p-dominant equilibrium [31]. In [32], a detailed study of 2×2 bimatrix coordination game using quantal response equilibrium as a homologous method of equilibrium selection is presented. Dreves proposes a new solution concept for

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the generalized NE problem, selecting a rational equilibrium from a typically infinite number of NEs [33]. Literature [34] reports an experiment examining behavioral and equilibrium selection in similar, infinitely repetitive games based on the Stag Hunt and the Prisoner's Dilemma. In [35] a method is demonstrated to fit the parameters of a continuous chicken model and to model pedestrian-pedestrian interactions in the laboratory. A chicken game based motion control algorithm is proposed for autonomous vehicles equipped with cooperative adaptive cruise control system in [36]. In addition, Some studies use the Lemke-Howson algorithm to compute the numerical solution of NE in the field of automated driving. Only a few of them discuss the existence of multiple NEs, and even fewer investigate how to select from mutiple NEs. Xu et al. uses the Lemke-Howson algorithm to find possible mixed strategies in the absence of pure Nash Equilibria (NEs) and does not mention the existence or treatment of multiple NEs [37]. Hu et al. presents the existence of multiple equilibria for the game-induced nonlinear opinion dynamics (GiNOD) problem but does not offer a clear solution [38]. Reference [39] proposes a theoretical method to analyze the uncertainty of potential equilibrium using Bayesian inference with an NE selection method, but it is not tested with human drivers. In summary, most of these equilibrium selection approaches have been studied by playing a game only once or repeating the same static game, and little research has been done on the balanced selection for autonomous driving dynamic games.

C. Contributions

To address the above issues in the existing literature, this paper proposes a decision-making framework based on dynamic game and NE selection at uncontrolled intersections. The interaction of participants in a traffic scenario is considered to enable AVs to pass through intersections safely and efficiently. The novelty lies in analyzing the problem of multiple NEs in the framework and proposing an effective mechanism for NE selection. This allows autonomous vehicles to get along better with their surroundings in the real world.

The contributions of our paper are as follows:

1) The problem of multiple closed-loop NEs is analyzed for uncontrolled intersection dynamic games.
2) A novel NE selection mechanism is designed based on the proposed new concept of CAB-Games to consider the potential conflicts at uncontrolled intersection.
3) A real-time game theoretic decision-making framework for AVs at uncontrolled intersections is proposed, which performs well in the interactions with other autonomous driving strategies and human drivers.

For the first time, the problem of multiple NEs are analyzed in the field of autonomous driving and a feasible solution is proposed to inspire future AI-based and date-driven decision-making approaches. The decision-making framework can also be used in more complex scenarios with high and diverse traffic density and multi-agent interaction.

The remainder of the paper is organized as follows. In Section II, an intersection scenario is modelled and a multi-factor cost function is designed. Section III proposes a game theoretic decision-making framework at uncontrolled intersections, considering the problem of closed-loop NE selection. Simulations and analysis are carried out in Section IV and Section V presents experiments and analysis using a driving simulator with human drivers. Finally, the conclusions and outlook are presented in Section VI.

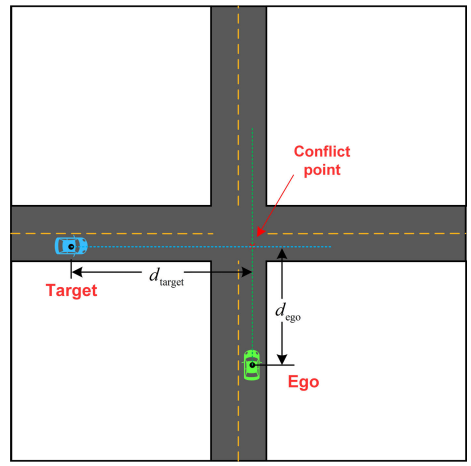

Fig. 1. Traffic scenario of two vehicles interacting at an uncontrolled intersection.

II. Problem Formulation

In this section, a scenario of two vehicles interacting at an uncontrolled intersection is established, as shown in Fig. 1.

A. Uncontrolled Intersection Setup

It is assumed that both the ego vehicle and the target vehicle move in a straight line, so only the longitudinal control (the throttle and brake) of the vehicles needs to be considered. The two vehicles need to make decisions individually, and the goal is to cross the intersection safely.

1) Vehicle Kinematic Model: To characterize the vehicle's state, a simplified discrete model is used to represent the vehicle's kinematics in this study.

$$\begin{aligned} d(k+1) &= d(k) - v(k)\cdot\Delta t - 0.5a(k)\cdot\Delta t^2 \\ v(k+1) &= v(k) + a(k)\cdot\Delta t, \end{aligned} \tag{1}$$

where d is the distance from the vehicle to the conflict point, v and a are the speed and acceleration of the vehicle moving in a straight line, respectively; k denotes the step and Δt is the sampling time period. In this model, $x(k) = [d(k), v(k)]^{\mathrm{T}}$ represents the state of the vehicle and $u(k) = a(k)$ represents the action of the vehicle.

2) Action Space: The actions are related to vehicle longitudinal motion. The vehicle's actuation unit consists of two parts: the upper controller determines the coarse action sequence, and the lower controller tracks the control commands from the upper controller [20]. Assuming that in the upper controller, the action space of AV consists of the following five discrete actions.

- "Maintain": maintain the speed, that is, $a = 0$.
- "Accelerate": Acceleration at $a = 1\mathrm{m/s}^2$.
- "Decelerate": Acceleration at $a = -1\mathrm{m/s}^2$.
- "Hard accelerate": Hard acceleration at $a = 2\mathrm{m/s}^2$.

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- "Hard decelerate": Hard deceleration at $a = -2\text{m/s}^2$.

In the dynamic game model, only the coarse upper controller is considered.

3) Constraints: The constraints of the model are desired restrictions on states or outputs as follows [40]:

$$\begin{aligned} v_{\min} \le v(k) \le v_{\max} \\ d_{\min} \le d(k) \le d_{\max}, \end{aligned} \tag{2}$$

where $v_{\min}$, $v_{\max}$, $d_{\min}$, $d_{\max}$ are hypothetical upper and lower bounds for the state in this scenario.

B. Cost Function Design

Driving is a multi-attribute task, which implies that it contains many attributes that have a significant impact (e.g., progress towards the destination, driving time, collisions, passenger experience, etc.). More specifically, the expectation is that AVs could cross intersections safely, while avoiding the inefficiencies resulting from overly conservative driving policies. In order to achieve safety, transportation efficiency and passenger comfort [41], the running cost function from the step k to $k+1$ in this study is designed as follows.

$$g_k(x_1, x_2, u_1, u_2) = w^{\mathrm{T}} j, \tag{3}$$

where x_1, x_2 and u_1, u_2 represent the states and actions from the step k to $k+1$ of two players; $w \in \mathbb{R}_+^3$ is a vector of weights and $j = (j_{safe}, j_{time}, j_{comfort})^{\mathrm{T}}$. These cost terms $j_{safe}, j_{time}, j_{comfort}$ are defined as follows [42]:

- j_{safe} represents the safety assessment of the vehicle, i.e. whether any collision has occurred. The simplified vehicle model is represented by a rectangular bounding box of $L_l \times L_w$. If the vehicle's bounding box overlaps another vehicle's bounding box, $j_{safe} = 1$; otherwise $j_{safe} = 0$. The weighting of j_{safe} is chosen to be very large to prioritize vehicle safety over other considerations.
- j_{time} represents the efficiency of the vehicle crossing the intersection, hence j_{time} is designed the remaining predicted passage time.

$$j_{time} = \frac{d(k+1)}{v(k+1)}. \tag{4}$$

- $j_{comfort}$ represents passenger comfort, which is assessed primarily by the acceleration and speed of the vehicle.

$$j_{comfort} = \theta a(k)^2 + (1-\theta)(v(k+1) - v_{ref})^2, \tag{5}$$

  where v_{ref} is the desired speed of the vehicle and θ is the weighting factor. Both parameters can be adjusted according to the preferences of drivers.

As a result, the running cost function can adjust its parameters to accommodate different drivers [43], [44].

When any vehicle passed through the intersection at the step k_{final}, the sum of all running costs is the total cost of this interaction as,

$$J = \sum_{k=0}^{k_{final}} g_k. \tag{6}$$

C. The Intersection Dynamic Game

In a general formal game, participants, policies and cost are the three basic elements. We can define a 2-player game as a tuple $(\mathcal{A}, \mathcal{J})$, where $\mathcal{A} = A_1 \times A_2$ is the set of actions available for player i, and $\mathcal{J} = J_1 \times J_2$, with J_i is the action-dependent cost functions of player i. A desirable game solution is one in which individual players act in accordance with their incentives to minimize their own cost. The concept of NE best embodies the thought and has become a core solution concept in game theory. Each player $i \in 1, 2$ chooses u_i to try to minimize the cost J_i. The closed-loop NE is one in which participants make decisions at the same time and can receive feedback to change their own control policies dynamically. A closed-loop NE is defined as any vector $(u_1, \cdots, u_N) \in \mathcal{A}$ such that [45]

$$J_i(u_1, \ldots, u_N) \le \inf_{\alpha \in A} J_i(u_1, \ldots, u_{i-1}, \alpha, u_{i+1}, \ldots, u_N). \tag{7}$$

In dynamic games, a player makes decisions sequentially and their states form a trajectory. Let $T_i = [x_i(0), x_i(1), \ldots]$ be the trajectory of player i. Extending the definition of NE to dynamic game, we can define a closed-loop NE of dynamic game as any trajectories set $(T_1, \cdots, T_N) \in \mathcal{T}$ if it satisfies

$$J_i(T_1, \ldots, T_N) \le \inf_{\beta \in \mathcal{T}} J_i(T_1, \ldots, T_{i-1}, \beta, T_{i+1}, \ldots, T_N). \tag{8}$$

where $\mathcal{T}$ is the set of trajectories available.

In the autonomous vehicle intersection dynamic game, when the state pair is $(x_1, x_2) = ([d_1, v_1]^T, [d_2, v_2]^T)$ and the action pair is (a_1, a_2), the state pair dynamics can be written as

$$(x_1^+, x_2^+) = f_{sys}((x_1, x_2), (a_1, a_2)), \tag{9}$$

based on the single vehicle kinematic model in (1). The running cost of the two vehicles can be written as

$$(l_1, l_2) = f_{rc}((x_1, x_2), (a_1, a_2)), \tag{10}$$

based on the single vehicle cost function in (3), where l_1 and l_2 is the running cost of Vehicle 1 and 2, respectively.

III. Game Theoretic Decision-Making Framework

In this section, a conflict-aware game theoretic decision-making framework for autonomous vehicles at uncontrolled intersections is proposed. Fig. 2 illustrates the framework. First, for each state, the closed-loop NE solutions of the dynamic game are calculated. We show that more than one closed-loop NEs can be found for certain states, and the selection of NEs is a nontrivial task. Then an NE selection mechanism is designed based on a series of sequentially-constructed static games, which we call the Conflict-Aware Bimatrix Games (CAB-Games). The idea of risk dominant equilibrium is used in the NE selection when there are multiple NEs in a CAB-Game. For most states, a deterministic action can be obtained after the closed-loop NE calculation and the NE selection, which are both conducted offline. For the remaining states, the action will be determined online after observing the other vehicle's behavior history. Finally, after a safety check, the vehicle executes the action.

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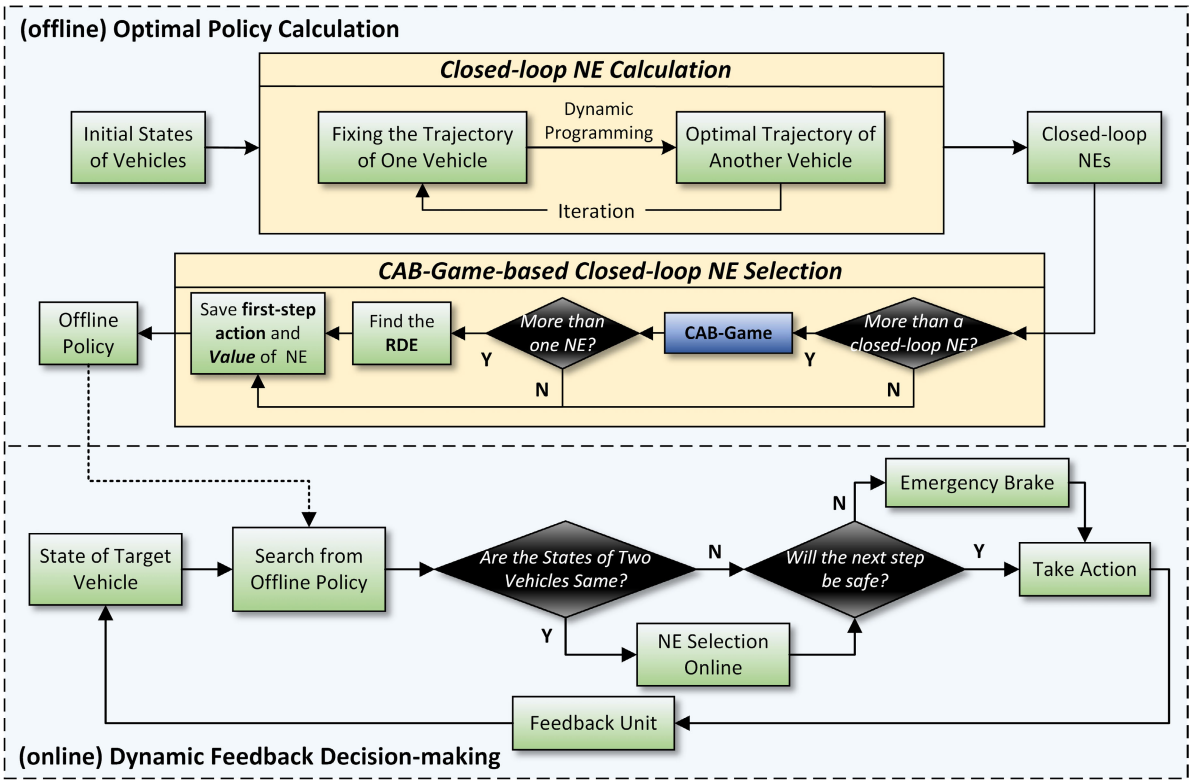

Fig. 2. Dynamic game theoretic decision-making framework.

A. Closed-Loop NE Solutions

1) Calculating the Closed-Loop NE Set: In this study, dynamic programming is used to calculate closed-loop NE solutions of dynamic games in an iterative fashion. Dynamic programming is a mathematical method to find the global optimal solution of a decision-making process with relatively low time and space complexity. The application of dynamic programming for discrete systems leads to the recurrence equation as,

$$J_t^*(x(k)) = \min_{u(k)}[g_t(x(k), u(k), k) + J_{t+1}^*(x(k+1))], \quad (11)$$

where g_t and J_{t+1}^* represent respectively the running cost from moment t to $t+1$ and minimum cost to reach the final cost to go.

For each pair of vehicle states, an initial trajectory of Vehicle 1 $T_1^0 = [x_1(0), x_1(1), \ldots]$ is first generated in some way (e.g., a keeping constant speed, or accelerating immediately). Then, fixing this trajectory of Vehicle 1, the optimal trajectory of Vehicle 2, denoted by T_2^1, is solved by dynamic programming. Next, the trajectory T_2^1 is fixed, and the optimal trajectory of Vehicle 1, denoted by T_1^1, is calculated using dynamic programming. The procedure is repeated, until $T_1^n = T_1^{n-1}$, that is, T_1^n and T_2^n are mutually optimal. The pair of mutually optimal trajectories is a closed-loop NE, while preserving the corresponding cost as the optimal cost R_1^*, R_2^*. With different initial trajectories of Vehicle 1, i.e., different T_1^0, this iterative computation may converge to different NEs.

Algorithm 1 presents a pseudocode for closed-loop NE calculation algorithm. In this pseudocode, j_m is the number of initial trajectories we try for each pair of states, which is set to 12 in this study. n_{max} is the maximum number of loop iterations, which is set to 10 in this study. This algorithm is applied to each state pair. As a result, closed-loop NEs U^* corresponding to each pair of vehicle states can be obtained.

Algorithm 1 Closed-Loop NE Calculation

input : Initial states $x_1(0), x_2(0)$
output: Closed-loop NE set $\mathcal{T}$ and optimal cost R_i^*

```
Let the closed-loop NE set be an empty set: T = φ;
for j = 1 to j_m do
    Generate an initial trajectory of Vehicle 1
    T_1^0 = [x_1(0), x_1(1), ...];
    for n = 1 to n_max do
        Fix T_1^{n-1}, calculate T_2^n and cost R_2^n by dynamic
        programming;
        Fix T_2^n, calculate T_1^n and cost R_1^n by dynamic
        programming;
        if Converge cyclically T_1^n == T_1^{n-1} then
            Update the closed-loop NE set:
            T ← T ⋂ {(T_1^n, T_2^n)};
            Update the optimal cost: R_i^* = R_i^n;
            break;
        end
    end
end
```

2) The Existence of Multiple NEs: For some vehicle state pairs, only one closed-loop NE can be found no matter how the initial trajectory changes. For some other vehicle state pairs, multiple NEs can be found when using different the initial

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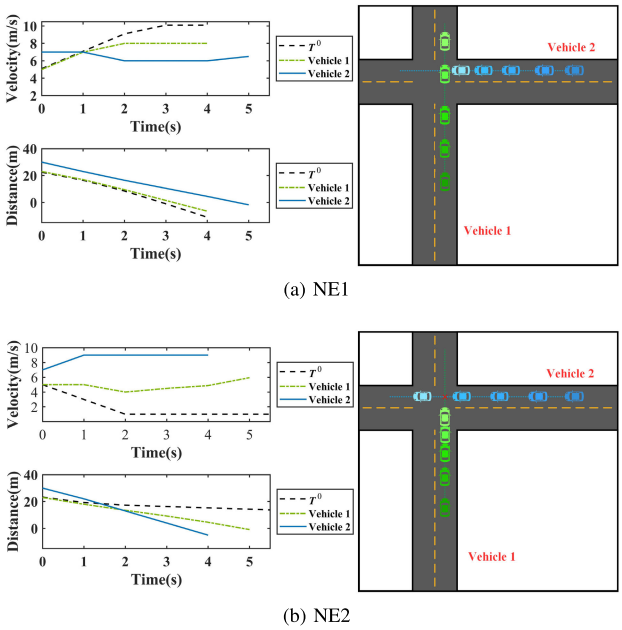

Fig. 3. An example of multiple closed-loop NEs: the initial states of the two vehicles is set to $x_1(0) = [23, 5]^{\mathrm{T}}$, $x_2(0) = [30, 7]^{\mathrm{T}}$. (a) When the initial trajectory of Vehicle 1 is set to accelerate first, the calculated NE1 shows that Vehicle 1 passes before Vehicle 2. (b) When the initial trajectory of Vehicle 1 is set to decelerate first, the calculated NE2 shows that Vehicle 1 passes after Vehicle 2.

trajectories in the iteration. It is obvious that when the state of the two vehicles are the same, there are two symmetric NEs. But multiple NEs may also occur when the states of the two vehicles are different. An example is given in Fig. 3 to illustrate a case with multiple NEs. The green solid and blue dotted lines represent the optimal trajectories of Vehicle 1 and Vehicle 2, respectively, and the black dotted line represents the initial trajectory of Vehicle 1 in the iteration. When the initial trajectory is set to accelerate first, the calculated closed-loop NE is shown in Fig. 3a. In this NE, Vehicle 1 behaves aggressively to pass through the intersection firstly. When the initial trajectory of vehicle 1 is set to decelerate first, a different closed-loop NE is found. Vehicle 1 behaves conservatively to pass through the intersection secondly in Fig. 3b. In the two closed-loop NEs, the two vehicles behave in an opposite way with a different passing order.

B. Closed-Loop NE Selection

Due to the presence of multiple NEs, closed-loop NE solutions in dynamic game can not be directly converted in to an implementable policy for AVs at intersections. Different NEs favor different players, and the two sides may clash if they don't agree. An NE selection mechanism considering conflict-aware is needed to filter out undesired NEs, so that the selected NEs are more likely to make both sides reach consensus. The key in the NE selection is to consider potential conflicts, that is, two vehicles selecting different NEs such that a new game will form at the next step. We propose a new concept called the Conflict-Aware Bimatrix Game (CAB-Game) to fully consider the potentially-conflicting situations in the design of autonomous vehicle policies.

TABLE I
THE ONE-STEP STATIC CONFLICT-AWARE BIMATRIX GAME

Vehicle 1 Action \ Vehicle 2 Action	a_{21} from NE1	a_{22} from NE2
a_{11} from NE1	Next-step state: $(x^+_{1,11}, x^+_{2,11})$ Vehicle 1 cost: $C_{1,11}$ Vehicle 2 cost: $C_{2,11}$	Next-step state: $(x^+_{1,12}, x^+_{2,12})$ Vehicle 1 cost: $C_{1,12}$ Vehicle 2 cost: $C_{2,12}$
a_{12} from NE2	Next-step state: $(x^+_{1,21}, x^+_{2,21})$ Vehicle 1 cost: $C_{1,21}$ Vehicle 2 cost: $C_{2,21}$	Next-step state: $(x^+_{1,22}, x^+_{2,22})$ Vehicle 1 cost: $C_{1,22}$ Vehicle 2 cost: $C_{2,22}$

1) Constructing CAB-Games: The basic idea behind our closed-loop NE selection mechanism is to assign a *value* to each state and construct a one-step static game for each decision-making using running cost and the value of the next-step state. This value, denoted as $V(x_1, x_2)$, is similar to the *cost-to-go* of an optimal control problem. It reflects how much one does not want to reach this state, which is initialized by the optimal cost R_i^* obtained in the previous section when calculating the closed-loop NE. Then, if there are multiple closed-loop NEs at the current state, a one-step static game can be constructed while assuming both players take actions to pursuit an NE, and they may choose the same NE or different NEs. For example, in one closed-loop NE, the actions of the two vehicles are a_{11} and a_{21}, respectively. In another closed-loop NE, the actions of the two vehicles are a_{12} and a_{22}, respectively. When the two vehicles chooses the actions a_{11} and a_{22}, respectively, they pursue different NEs. Let $(x^+_{1,jk}, x^+_{2,jk})$ denote the next-step state, when the two vehicles take actions a_{1j} and a_{2k} respectively, $j, k \in \{1, 2\}$(When we calculate the closed-loop NEs, we find that there are no more than two closed-loop NEs in the dynamic game, which is generally true for interaction games, but this one-step static game construction method can be generalized to cases with more than two NEs). Let $V^+_{i,jk}$ denote the value of Vehicle i at the next-step state and $l_{i,jk}$ denote the running cost of Vehicle i, when the two vehicles take actions a_{1j} and a_{2k} respectively, $i, j, k \in \{1, 2\}$. The cost of this one-step static game for Vehicle i is determined by the sum of the one-step running cost and the value of the next state, that is,

$$C_{i,jk} = l_{i,jk} + V^+_{i,jk}, \quad i, j, k \in \{1, 2\}, \tag{12}$$

and the game is shown in Table I. If there are only a finite number of closed-loop NEs in the two vehicle dynamic game, this one-step static game is always a bimatrix game. We call this one-step static game the Conflict-Aware Bimatrix Game (CAB-Game). Based on this CAB-Game, an action is selected and the value of the current state is determined (more details in Section III-B.2). The NE selection of each state is performed in the order of the distance d from smallest to largest, similarly to a backward induction process. This guarantees that the corresponding value of the next state is known when we construct a CAB-Game.

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2) RDE-Based CAB-Game NE Selection: CAB-Games are constructed for state pairs with two NEs. Each CAB-Game itself may have one or two NEs. If it has only one NE, this NE can be selected as the autonomous vehicle policy. If it has two NEs, we propose a selection method based on the idea of risk dominant equilibrium (RDE).

In a general 2×2 bimatrix game, both players have two actions, namely, (A_1, A_2) and (B_1, B_2), to choose from. When the two players choose the actions A_i and B_j, the cost of the two players are α_{ij} and β_{ij}, respectively. The cost matrix can be written as

$$\begin{array}{c} \\ A_1 \\ A_2 \end{array} \begin{array}{c} \begin{array}{cc} B_1 & B_2 \end{array} \\ \begin{pmatrix} \alpha_{11}, \beta_{11} & \alpha_{12}, \beta_{12} \\ \alpha_{21}, \beta_{21} & \alpha_{22}, \beta_{22} \end{pmatrix} \end{array}. \tag{13}$$

NEs for bimatrix games with a finite number of participants and actions can be calculated using the Lemke-Howson algorithm [46]. The risk dominant equilibrium, proposed by Harsanyi and Selten in 1988 [29], is the NE with a maximum Nash product. The Nash product is the product of the deviation losses of the two participants at a particular equilibrium. The RDE is more attractive to players when they are uncertain about the actions of other players. In the bimatrix game, the action set (A_1, B_1) is called risk dominant if the Nash product satisfies [9]

$$(\alpha_{11} - \alpha_{21})(\beta_{11} - \beta_{12}) > (\alpha_{22} - \alpha_{12})(\beta_{22} - \beta_{21}). \tag{14}$$

If the CAB-Game has two NEs, the RDE is selected as it is considered as more attractive when one vehicle is uncertain about the strategy of the other vehicle.

3) CAB-Game-Based Closed-Loop NE Selection: Combining the CAB-Game construction and the RDE-based CAB-Game NE selection, we obtain an NE selection method for the dynamic game. This method can be considered as an NE selection mechanism for the dynamic game, because the resultant policy always chooses an action of a closed-loop dynamic game NE. The actions of CAB-Game are the same as the first-step actions of the original dynamic game by design. Therefore, this approach indeed selects a closed-loop NE for the dynamic game at every step. The CAB-Game-based closed-loop NE selection algorithm is shown in Algorithm 2. The closed-loop NE selection is reverse normalized based on the ascending order of the displacements in the vehicle state pairs. When displacements are small there is often only one closed loop NE due to the presence of collision constraints. The backward induction approach enables the NE selection process to find the unique Value corresponding to each next state in the cost matrix for each selection.

The proposed algorithm works for almost all state pairs, except the symmetric state pairs, where there is no RDE in the CAB-Game as the two NEs are symmetric. When the two vehicles are in the same state $x_1 = x_2$ with the same cost function, the two CAB-Game NEs are symmetric. It is difficult to compare them and make decisions in the offline process. Then, the selection of NEs in this case will be determined in online, where history driving behavior of the other vehicle is considered. A value of the state pair is still estimated offline so the backward induction process can continue. The action selection of this state pair is marked as to-be-determined and it will be decided online.

Ultimately, the *Value* corresponding to each initial state can be obtained to ensure that a unique optimal action can be searched from the offline set for each online decision.

C. Online Decision-Making

The online decision-making module has two tasks: (1) determining the actions for symmetric states where no deterministic decisions can be made offline (no RDE exists in its CAB-Game therefore its action is to-be-determined), and (2) evaluating the safety risk and applying emergency braking when necessary.

When the state of the two vehicles are the same, there are two symmetric NEs. The corresponding actions of the two symmetric NEs are a^*_{slow} and a^*_{fast}, where $a^*_{slow} < a^*_{fast}$. The online decision is made based on the 3-step history of the other vehicle: if the other vehicle is aggressive, the decision is to be conservative, i.e., $u = a^*_{slow}$; otherwise, the decision is to be aggressive, i.e., $u = a^*_{fast}$. We use the sum of the 3-step history acceleration values to indicate if the other vehicle is aggressive or conservative.

Finally, the optimal policy obtained is checked for safety. The trajectories of the two vehicles in the next three steps (x_1^+, x_2^+) is predicted, based on the kinematic model of the vehicles in (1), the current states and history information of the vehicles. If there is a safety risk, emergency braking is applied, that is, $u_1^*(k) = u_{Emer}$, where $u_{Emer} = -5\text{m/s}^2$ is the acceleration of emergency braking. In this way, the autonomous vehicles search for the optimal policy to execute from the offline policy set in real time according to the observed states of the two vehicles in the interaction, where the time spent is almost negligible.

Algorithm 3 presents a pseudocode for the online dynamic feedback decision-making algorithm.

IV. Simulations

In this section, simulations are carried out to demonstrate the effectiveness of the game theoretic decision-making algorithm. The uncontrolled intersection in Fig. 1 is considered, where vehicles follow the intersecting trajectories. Firstly, the offline policy sets are calculated offline. Then a simulation study is conducted to test the self-consistency (interaction between AVs with the proposed policy) and interactivity (interaction between AV with the proposed policy and vehicles with other policies) of the algorithm proposed in this study.

A. Parameter Settings and Offline Calculation Results

Since this study targets the decision-making framework for low-speed vehicle operations at uncontrolled intersections, the parameters in the scenarios are set as

$$\begin{cases} v_{min} = 0, v_{max} = 10\text{m/s} \\ d_{min} = -10\text{m}, d_{max} = 40\text{m}, \end{cases} \tag{15}$$

where d_{min} is set to a negative number so that the vehicles can pass completely through the intersection. The acceleration

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Algorithm 2 CAB-Game-Based Closed-Loop NE Selection

input : Closed-loop NEs $\mathcal{T}$ and optimal cost R_i^*
output: Policies $\pi_i(x_1, x_2)$ and values $V_i(x_1, x_2)$ for all valid state pairs (x_1, x_2), $i \in \{1, 2\}$

Values initialization: $V_i(x_1, x_2) = R_i^*((x_1, x_2))$;
for (x_1, x_2) *in ascending order of the distance* d **do**
  if *two NEs in* $\mathcal{T}$ *starting at* (x_1, x_2) **then**
    Find the current-step actions of the two NEs: a_{11}, a_{12} of Vehicle 1, and a_{21}, a_{22} of Vehicle 2;
    Calculate potential state pairs at the next step: $(x_{1,jk}^+, x_{2,jk}^+) = f_{sys}((x_1, x_2), (a_{1j}, a_{2k}))$, $j, k \in \{1, 2\}$;
    Calculate potential running costs: $(l_{1,jk}, l_{2,jk}) = f_{rc}((x_1, x_2), (a_{1j}, a_{2k}))$, $j, k \in \{1, 2\}$;
    Find values of potential next-step state pairs: $V_{1,jk}^+ = V_1(x_{1,jk}^+, x_{2,jk}^+)$, $V_{2,jk}^+ = V_2(x_{1,jk}^+, x_{2,jk}^+)$, $j, k \in \{1, 2\}$;
    Calculate total cost as the sum of the running cost and the next-step value: $C_{1,jk} = l_{1,jk} + V_{1,jk}^+$, $C_{2,jk} = l_{2,jk} + V_{2,jk}^+$, $j, k \in \{1, 2\}$;
    Construct a bimatrix game (the CAB-Game): $\begin{array}{c} \\ a_{11} \\ a_{12} \end{array} \begin{array}{c} \begin{array}{cc} a_{21} & a_{22} \end{array} \\ \begin{pmatrix} C_{1,11}, C_{2,11} & C_{1,12}, C_{2,12} \\ C_{1,21}, C_{2,21} & C_{1,22}, C_{2,22} \end{pmatrix} \end{array}$;
    if *two NEs in the bimatrix game* **then**
      Find the RDE of the bimatrix game;
      Update the policy: $\pi_i(x_1, x_2)$ = the action of the bimatrix game RDE, $i \in \{1, 2\}$;
      Update the value: $V_i(x_1, x_2)$ = the cost of the bimatrix game RDE, $i \in \{1, 2\}$;
    end
    else if *only one NE in the bimatrix game* **then**
      Update the policy: $\pi_i(x_1, x_2)$ = the action of the bimatrix game NE, $i \in \{1, 2\}$;
      Update the value: $V_i(x_1, x_2)$ = the cost of the bimatrix game NE, $i \in \{1, 2\}$;
    end
  end
  else if *only one NE in* $\mathcal{T}$ *starting at* (x_1, x_2) **then**
    Update the policy: $\pi_i(x_1, x_2)$ = the current-step action of the closed-loop NE, $i \in \{1, 2\}$;
    Update the value: $V_i(x_1, x_2)$ = the cost of the closed-loop NE, $i \in \{1, 2\}$;
  end
end

values are set as the actions in Section II-A.2. It is assumed that the parameters of the vehicle model are $L_l = 4.5$m, $L_w = 2$m, and the parameters of the cost function are given as

$$\begin{cases} w = (10^6, 20, 2.5)^{\mathrm{T}} \\ \theta = 0.8 \\ v_{ref} = 6\mathrm{m/s} \\ C = 3\mathrm{m/s}^2. \end{cases} \tag{16}$$

When calculating the offline policy using dynamic programming, each set of initial states consists of four parameters (d_1, v_1, d_2, v_2). The state values are quantized into a four-dimensional grid. Considering the calculation accuracy and calculation volume, the quantization interval of distance and speed is set to $\Delta d = 1$m and $\Delta v = 1$m/s. In our method, the average offline computation time for each vehicle state pair is about 1 second, and all the offline calculation took a total of 40 hours on a laptop. The time can be reduced if the calculations are performed on a workstation.

The game-theoretic calculation results are summarized in Table II. For 9.5% of the initial states, vehicles will definitely collide because they are too fast or the initial distance is too close. Note that it is not that the closed-loop NE is not found but that there is a collision in the calculated NEs. Emergency braking can ensure as much safety as possible during simulations and experiments. For 75.1% of the initial states, there is only one closed-loop NE in the dynamic game. For the remaining 15.4%, there are two closed-loop NEs. Among those initial states with two closed-loop NEs, most have one pure NE in the constructed CAB-Game, therefore, the choice of their NE can be determined by the action of the CAB-Game NE. A small portion have multiple NEs in the CAB-Game, but their NE can be selected based on RDE. Less than 1% have a pair of symmetric NEs as the states of the two vehicles are the same, and less than 1% have one mixed NE.

B. Self-Consistency Simulations

In this subsection, the self-consistency of the dynamic game-theoretic decision-making (DGTD) policy proposed in this paper is verified, where both vehicles at the intersection use the same policy.

We choose the MPC, more relevant level-k theory for comparison on the results with the self-interaction [47]. Since the observed sophistication of human strategists is barely beyond level-2 [48], level-1 and level-2 policies be used in the simulation. In the MPC policy, it is assumed that the target vehicle maintains its current speed and the model predictive control is used to find the local (next five steps) optimum for the ego vehicle as the optimal policy. According to the level-k

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Algorithm 3 Online Decision-Making

input : Real-time states $x_1(k), x_2(k)$ and action history $u_2(k-1), u_2(k-2), u_2(k-3)$
output: Optimal action $u_1^*(k)$

Obtain states of two vehicles: $(x_1(k), x_2(k))$;
Find the NEs from the offline policy after CAB-Game and RDE-based selection;
if *there are still two NEs (the two vehicle states are the same)* **then**
  Calculate the sum of historical actions of other vehicle: $S = u_2(k-3) + u_2(k-2) + u_2(k-1)$;
  if $S \geq C$, *where C is a threshold* **then**
    Update the action: $u_1(k) = a^*_{slow}$;
  else
    Update the action: $u_1(k) = a^*_{fast}$;
  end
end
else if *there is one selected NE* **then**
  Update the action: $u_1(k) = a^*$;
end
Predict the next three step of two vehicles;
if *No safety risk in next three states* **then**
  Update the optimal action: $u_1(k) = u_{Emer}$;
else
  Update the optimal action: $u_1(k) = u_1(k)$;
end
Optimal action $u_1^*(k) \leftarrow u_1(k)$

TABLE II
RESULTS OF OFFLINE CLOSED-LOOP NE CALCULATION

Dynamic game	CAB-Game	Number of states	Percentage (%)
Collision	–	29859	9.5
One closed-loop NE	–	236496	75.1
Two closed-loop NEs	Symmetric states	230	<1
	One pure NE	41405	13.2
	One mixed NE	405	<1
	One RDE	6326	2.0
Total		314721	100.0

game theory algorithm [23], [49], a level-0 vehicle moves at a constant speed without considering its surroundings or other vehicles. A level-k vehicle assumes all the other vehicles are with level-$(k-1)$ policies. Offline policies are calculated by dynamic programming after predicting the sequence of applied actions of the other vehicle. An emergency braking mechanism is set up for all policies to handle the emergency scenarios where both vehicles choose to go aggressively while assuming the other one would yield.

In the simulation, the state of the vehicle is not necessarily on the state grid of the offline sets at the time of decision-making, and the action may not be found directly from the offline policy sets. For this reason, interpolated searching is used in dynamic feedback decision-making. In the simulation, it is assumed that the lower-level control sampling frequency is 100Hz and the higher-level decision frequency is 2Hz, i.e. the sampling interval ΔT is 0.01s and the decision interval ΔT_{dec} is 0.5s. The number of simulations is set to 10,000 with random initial states sampled uniformly from the grid. All four policies are tested using the same 10,000 initial states.

TABLE III
RESULTS OF SELF-CONSISTENCY SIMULATION

Policy	Emergency braking rate (%)	Collision rate (%)	Average cost
MPC	10.5	1.96	524
Level-1	10.1	1.92	494
Level-2	24.6	4.00	394
DGTD	18.8	2.23	412

The self-consistent simulation results of different policies are shown in Table III. The emergency braking rate refers to the percentage of cases that emergency braking is triggered. Collision rate refers to the percentage of cases that the two AVs collide after the applied emergency braking. Some of the collisions are unavoidable due to its initial state. It should be noted that as we are comparing the higher level policies, the emergency braking mechanism in this study is applied at the policy level at 2 Hz, and all vehicles follow a pre-defined path without steering in simulation. Therefore, some of the collisions may be avoided by applying a lower-level emergency braking control at a higher frequency or allowing lateral movement of the vehicles. The emergency braking is set to a fixed value of -5m/s^2. Some collisions may also be avoided if a larger deceleration is used. The average cost indicates the cost of vehicles passing through the intersection without collision or emergency braking.

Both the MPC policy and the level-1 policy performs poorly in terms of its average cost. The level-2 policy, despite showing a low average cost, has the highest emergency braking rate and the highest collision rate. This implies that the level-2 policy is aggressive in terms of lowering its own cost. However, it only works out in some cases, as being aggressive may lead to unwanted results when the other vehicle tries to do the same. The proposed DGTD policy achieves a balance between the emergency braking/collision rates and the average cost. Its average cost is 17% lower than that of the level-1 policy, and its emergency braking and collision rates are also significantly lower than those from the level-2 policy.

C. Interactivity Simulations

In this subsection, two vehicles at the intersection use different policies. All other simulation settings are the same as those in the self-consistency simulation. The results of the interactivity simulation for these policies are presented in Table IV.

In general, the level-2 and the proposed DGTD policies have lower costs than the MPC and level-1 policies. The emergency braking rate and the collision rate of all policy pairs are similar, except those between the level-2 policy and the DGTD policy.

D. Comprehensive Comparison

Table. V summarizes the emergency braking rates and average costs of all simulations results.

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TABLE IV
RESULTS OF INTERACTIVITY SIMULATION

Policy		EB rate[1] (%)	Colli rate[2] (%)	Average cost	
Vehicle 1	Vehicle 2			Vehicle 1	Vehicle 2
MPC	Level-1	10.2	1.92	488	508
	Level-2	12.6	1.73	455	394
	DGTD	12.4	1.89	478	412
Level-1	MPC	10.2	1.92	508	488
	Level-2	12.6	1.71	455	395
	DGTD	12.2	1.83	479	409
Level-2	MPC	12.6	1.73	394	455
	Level-1	12.6	1.71	395	455
	DGTD	21.3	3.12	413	433
DGTD	MPC	12.4	1.89	412	478
	Level-1	12.2	1.83	409	479
	Level-2	21.3	3.12	433	413

[1] EB rate: Emergency braking rate
[2] Colli rate: Collision rate

TABLE V
RESULTS OF SIMULATION

Policy	EB rate[1](%)		Average cost		
	Self[2]	Inter[3]	Self	Inter (Ego)	Inter (Target)
MPC	10.5	11.7	524	473	438
Level-1	10.1	11.7	494	481	431
Level-2	24.6	15.5	394	401	448
DGTD	18.8	15.3	412	418	457

[1] EB rate: Emergency braking rate
[2] Self: Self-consistency Simulations
[3] Inter: Interactivity Simulations

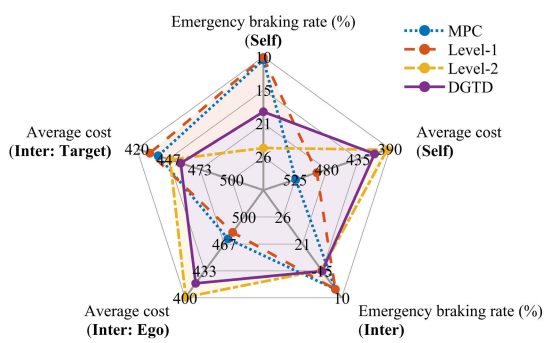

Fig. 4. Policies evaluation of simulation.

Five metrics from the simulation results were selected for evaluation, as shown in Fig 4. These five metrics are as follows:

1) the emergency braking rate when interacting against itself;
2) the average cost when interacting against itself;
3) the emergency braking rate when interacting against the other three policies;
4) the average cost of the ego vehicle when interacting against the other three policies;
5) the average cost of the target vehicle when interacting against the other three policies.

Fig. 5. Human driving simulator experiment platform.

As can be seen, MPC and level-1 policies perform more conservatively in self-consistency and interactivity simulations, thus performing well in emergency braking rate and also lead to larger cost. Level-2 policy performs best on interactive simulations, but contributes to the highest emergency braking rate when interacting with itself. However, due to the low cost characteristic of level-2 policy, there is an increasing percentage of AVs with level-2 policy in traffic scenarios. Further, it is easy to trigger safety emergency braking, due to the aggressive characteristics of level-2, which will affect traffic efficiency conversely. Therefore, the DGTD policy, providing a better balance between safety and efficiency, is more applicable to AVs in intersection interactions.

V. EXPERIMENTS

In this section, experiments are carried out to demonstrate the effectiveness of the game theoretic decision-making algorithm. Several existing studies have considered human preferences for AVs' decision making through driving simulator experiments [50], [51], [52]. Human-driven vehicles (HVs) interact with autonomous vehicles (AVs) using the proposed algorithm at uncontrolled intersections in a simulated environment. A driving simulator based on the simulation software CARLA is used in the experiment, as shown in Fig. 5. In the driving simulator experiment, the human driver has full control of the vehicle, including the pedals and the steering wheels, so this driving simulator experiment can reflect the real-world autonomous vehicle and human-driven vehicle interactions.

A. Experiment Design

An urban simulation scenario as presented in Fig. 6 was constructed using the software RoadRunner from MathWorks. The HV interacts with the AVs at four uncontrolled intersections, which are highlighted by the red boxes in the figure. The HV drives on the main road in the middle of the map, crossing four intersections each way. The AVs run on horizontal roads. The HV interacts with the AVs at eight times per lap.

Eight volunteer drivers with a driving license for at least 1 year are enrolled in the experiment. Each subject conducted

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TABLE VI
DRIVING SIMULATOR EXPERIMENT RESULTS

Policy	Volunteer driver #	Success rate (%)	AV EB rate[1] (%)	Collision rate[2] (%)	Number of successful cases	Normalized cost				APF rate[4] (%)
						HV mean	HV std[3]	AV mean	AV std	
Level-2	1	87.5	12.5	0	7	1.21	0.74	0.73	0.21	37.5
	2	75.0	25.0	0	6	1.88	1.01	1.25	0.49	25.0
	3	50.0	50.0	0	4	1.13	0.41	0.97	0.27	50.0
	4	25.0	75.0	12.5	2	1.81	1.46	0.58	0.02	75.0
	5	75.0	25.0	12.5	6	1.52	0.44	1.02	0.65	37.5
	6	75.0	25.0	0	6	1.22	0.56	1.22	0.40	50.0
	7	50.0	50.0	0	4	2.01	0.89	1.27	0.67	62.5
	8	62.5	37.5	12.5	5	0.70	0.31	1.32	1.46	50.0
	Overall	**62.50**	**37.50**	**4.69**	40	**1.44**	0.73	**1.05**	0.52	48.44
DGTD	1	62.5	37.5	0	6	1.85	0.54	0.96	0.28	37.5
	2	100.0	0.0	0	8	1.35	0.37	1.16	0.39	50.0
	3	50.0	50.0	0	4	1.60	0.55	0.89	0.28	50.0
	4	75.0	25.0	0	6	1.64	0.62	0.97	0.51	25.0
	5	100.0	0.0	0	8	1.20	0.44	0.98	0.20	25.0
	6	87.5	12.5	0	7	1.54	0.61	0.96	0.30	50.0
	7	100.0	0.0	0	8	1.52	0.41	1.65	0.67	62.5
	8	75.0	25.0	0	6	1.15	0.20	0.97	0.20	37.5
	Overall	**81.25**	**18.75**	**0**	53	**1.48**	0.47	**1.07**	0.35	42.19

[1] AV EB rate: AV emergency braking rate. AV EB rate + Success rate = 100%.
[2] The collision cases are a subset of the emergency braking cases.
[3] std: standard deviation.
[4] APF rate: AV passing-first rate.

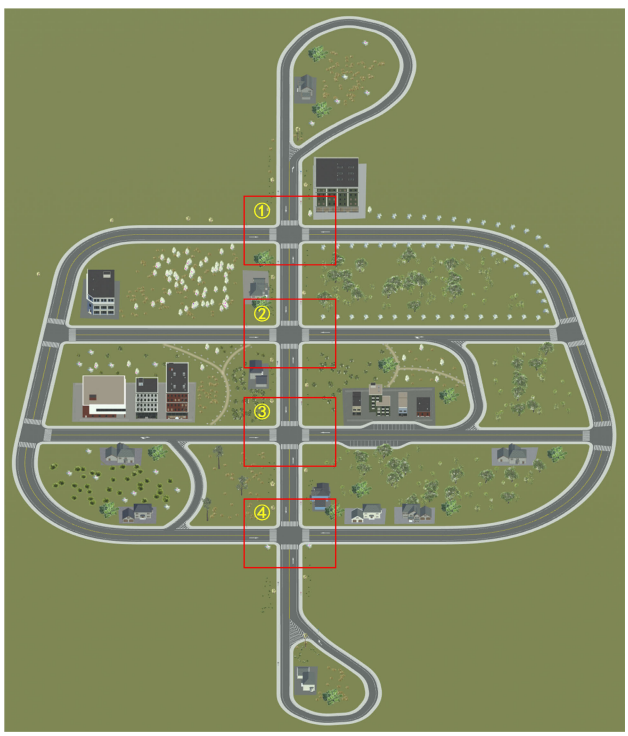

Fig. 6. The driving simulator scenario in RoadRunner.

two sets of experiments. In each set of experiments, the subject drove the HV for one lap. Since the level-2 and DGTD policies show significantly lower cost in the simulation than the other two policies, AVs implementing these two policies were evaluated with human drivers in the experiment. When the HV approaches the intersection, the initial states of the AVs are randomly generated each time. As in the simulation, the low-level tracking control frequency and high-level decision frequency of the vehicle were set to 100Hz and 2Hz.

B. Results Analysis

The experiment results are shown in Table VI. If the AV passes through the intersection without triggering the emergency brake during an interaction, it is considered as a *successful* case, otherwise it is recorded as an emergency braking case. The overall success rate of the DGTD policy is 81.25%, much higher than the level-2 policy at 62.50%. It is worth noting that three cases end up in a collision even after the AV emergency braking in the level-2 policy interaction. In the DGTD policy, there are no collisions. Since the initial state of the two vehicles in every interaction is random, the cost of each interaction is normalized using

$$c_{norm} = \frac{c_{real}}{c_{optimal}}, \tag{17}$$

where c_{real} is the cost in the experiment, and $c_{optimal}$ is the cost from the offline policy (see Fig. 2) of this initial state. The normalized cost mean and stand deviation values in Table VI only considers the passes without emergency braking, which include 40 cases for the level-2 policy and 53 cases for the DGTD policy.

Fig. 7 visualizes the summarized experiment results. As can be seen from Fig. 7a, the emergency braking rate of AV under DGTD policy is less than half the level-2 policy. Fig. 7b shows that the mean cost of HVs and AVs in the non-emergency-braking cases are similar for these two policies. We conduct hypothesis tests on the experiment data, and the results are shown in Table VII. The two proportions testing on the emergency braking rate data has a P-value less than the significance level of 0.05. Therefore, the alternative hypothesis H_a is accepted, i.e. the proposed DGTD policy has a lower emergency braking rate than the level-2 policy. The mean and variance of the normalized cost on successful cases are also

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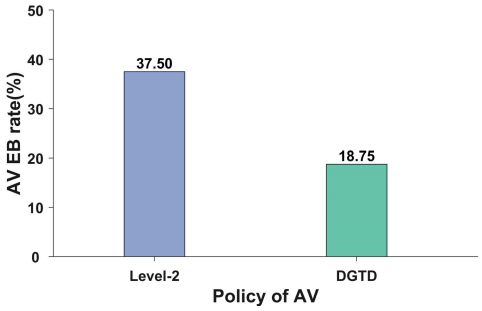
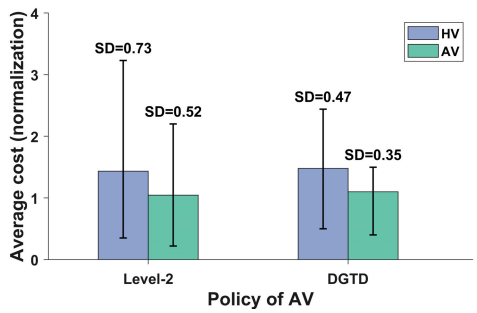


(a) Emergency braking (EB) rate of AV under two policies



(b) Cost analysis of AVs under two policies: average normalized cost and standard deviation (std) of successful samples

Fig. 7. AV EB rate and Cost analysis of driving simulator experiments.

TABLE VII
STATISTICAL TESTS ON EXPERIMENT RESULTS

Indicators	Test methods	H_0	H_a	*P*-values	
				HV	AV
AV emergency braking rate	Two proportions test	$p_1 = p_2$[1]	$p_1 > p_2$	—	**0.042**
Normalized cost mean	Two-sample *t*-test	$\mu_1 = \mu_2$	$\mu_1 \neq \mu_2$	0.733	0.895
Normalized cost variance	Bonett two variance test	$\sigma_1 = \sigma_2$	$\sigma_1 \neq \sigma_2$	0.316	0.620

[1] The 1 and 2 in the indicator subscripts refer to level-2 policy and DGTD policy, respectively.

tested. The *P*-value are all larger than the significance level of 0.05, so the null hypothesis H_0 cannot be rejected, i.e., there is no significant difference between the DGTD and the level-2 policies in terms of the normalized cost for either the HV or the AV.

Therefore, we conclude that the AV with the DGTD performs better than the level-2 policy in the experiment in terms of the emergency braking rate, and the cost performance on the successful cases are similar for the two policies. Since the DGTD policy integrates an NE selection mechanism that considers future states, it is able to behave smoothly without been too aggressive or too conservative.

C. Example Case Analysis

Three typical cases of vehicle intersection game with AV running the DGTD policy are selected from the driving simulator experiment for detailed demonstration.

TABLE VIII
VEHICLE INTERACTION DETAILS FOR CASE 1

Time(s)	States of vehicles	Online NEs selection	Optimal policy	Trend of AV
①t=0	$x_{av} = [36.6, 5.9]^{\mathrm{T}}$ $x_{hv} = [38.8, 7.3]^{\mathrm{T}}$	No	u_{av}=1	first pass
②t=1.5	$x_{av} = [27.2, 7.0]^{\mathrm{T}}$ $x_{hv} = [27.4, 7.2]^{\mathrm{T}}$	Yes	u_{av}=-2	later pass
③t=3	$x_{av} = [19.5, 5.2]^{\mathrm{T}}$ $x_{hv} = [17.6, 6.4]^{\mathrm{T}}$	No	u_{av}=-1.6	first pass
④t=4.5	$x_{av} = [14.0, 2.6]^{\mathrm{T}}$ $x_{hv} = [7.7, 6.9]^{\mathrm{T}}$	No	u_{av}=-1.9	later pass
⑤t=6	$x_{av} = [11.6, 6.9]^{\mathrm{T}}$ $x_{hv} = [-3.2, 7.7]^{\mathrm{T}}$	No	u_{av}=-0.9	later pass

TABLE IX
VEHICLE INTERACTION DETAILS FOR CASE 2

Time(s)	States of vehicles	Online NEs selection	Optimal policy	Trend of AV
①t=0	$x_{av} = [35.5, 8.9]^{\mathrm{T}}$ $x_{hv} = [37.0, 8.8]^{\mathrm{T}}$	No	u_{av}=0.2	first pass
②t=1	$x_{av} = [26.2, 9.5]^{\mathrm{T}}$ $x_{hv} = [28.0, 8.8]^{\mathrm{T}}$	No	u_{av}=0	first pass
③t=2	$x_{av} = [17.4, 9.0]^{\mathrm{T}}$ $x_{hv} = [19.8, 8.9]^{\mathrm{T}}$	No	u_{av}=-5	later pass
④t=3.5	$x_{av} = [8.6, 3.7]^{\mathrm{T}}$ $x_{hv} = [6.1, 9.0]^{\mathrm{T}}$	No	u_{av}=-1.5	later pass
⑤t=5	$x_{av} = [3.6, 3.6]^{\mathrm{T}}$ $x_{hv} = [-7.5, 9.2]^{\mathrm{T}}$	No	u_{av}=2	later pass

1) Case 1: **(Online NE selection.)** An example showing the online NE selection is shown in Fig. 8. The blue vehicle is the AV and the green vehicle is the HV in Fig. 8a. The points on the velocity and displacement curves in Fig. 8b are the AV decision times. The red markers mean the AV's strategic tendency to pass first and black ones represent passing later. In particular, the diamond dot represents policy through online NE selection. Furthermore, five decision-making times were selected and the vehicle interaction details for Scenario 1 are shown in Table VIII. The above analysis can indicate that the proposed policy has the flexibility to adjust the tendency according to the dynamic scenarios.

2) Case 2: **(A Changed Tendency and Emergency braking of the AV.)** Another example is shown in Fig. 9, where the triangles in Fig. 9b represent emergency braking policy. Likewise, five decision-making times were selected and the vehicle interaction details for Scenario 2 are shown in Table IX. The AV tried to pass first, but yielded later.

3) Case 3: **(Braking of the HV leads to a changed tendency.)** Another example is shown in Fig. 10, and five decision-making times were selected and the vehicle interaction details for Case 3 are shown in Table X. The AV decided to yield first. Then, as the HV also slowed down,

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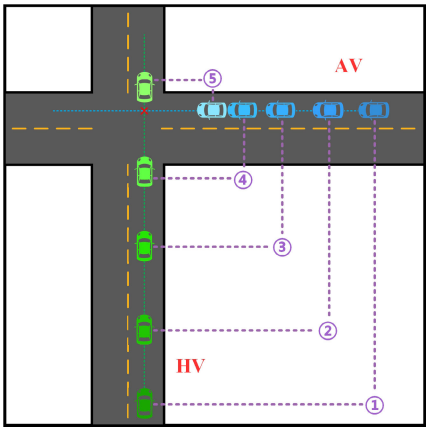
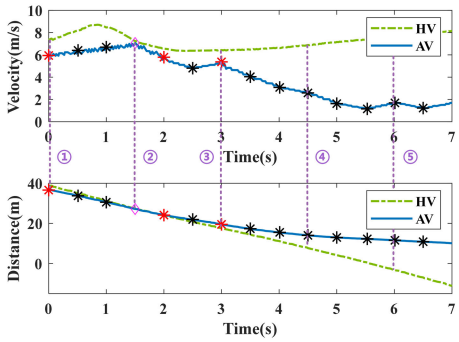


(a)



(b)

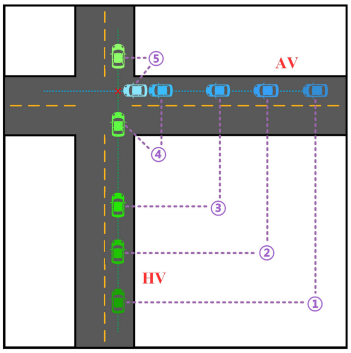
Fig. 8. Case 1: the HV eventually has priority through the intersection. The states of the two vehicles were similar at t = 3s $x_{av} \approx x_{hv}$, leading to an online closed-loop NE selection. The sum of the absolute values of the historical actions of HV was calculated as $\sum_{j=0}^{2} |u_{hv}(j/2)| = 4.1 > C$, thus the HV driver was considered aggressive driving styles and the AV implemented a conservative policy $u_{av} = -2$.

TABLE X
VEHICLE INTERACTION DETAILS FOR CASE 3

Time(s)	States of vehicles	Online NEs selection	Optimal policy	Trend of AV
①t=0	$x_{av} = [22.9, 7.0]^{\mathrm{T}}$ $x_h v = [27.5, 8.2]^{\mathrm{T}}$	No	u_{av}=-0.7	later pass
②t=1	$x_{av} = [16.3, 5.9]^{\mathrm{T}}$ $x_h v = [19.8, 7.1]^{\mathrm{T}}$	No	u_{av}=-1.6	later pass
③t=2.5	$x_{av} = [9.7, 3.3]^{\mathrm{T}}$ $x_h v = [12.0, 3.6]^{\mathrm{T}}$	No	u_{av}=2	first pass
④t=4	$x_{av} = [2.8, 5.4]^{\mathrm{T}}$ $x_h v = [8.1, 1.9]^{\mathrm{T}}$	No	u_{av}=1	first pass
⑤t=5.5	$x_{av} = [-6.0, 6.2]^{\mathrm{T}}$ $x_h v = [4.1, 3.9]^{\mathrm{T}}$	No	u_{av}=0.2	first pass

the AV accelerated to pass first. It can be seen that the AV can react quickly to real-time situations, which not only improves driving efficiency, but also matches human driving habits.

It can be summarized that the DGTD policy performs best in terms of balancing safety and efficiency in the experiment.



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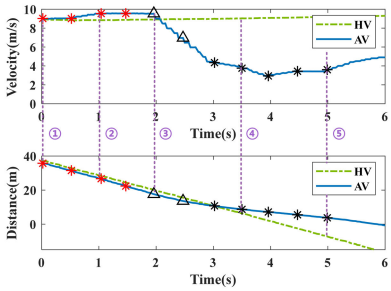
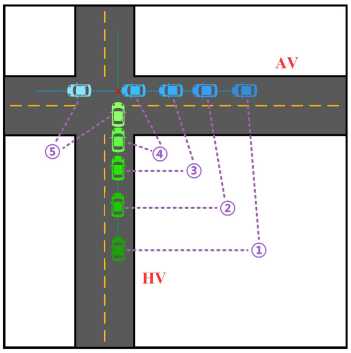
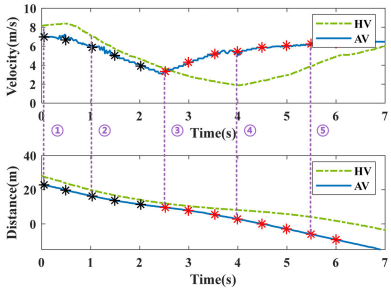


(b)

Fig. 9. Case 2: the HV passed through the intersection preferentially and the AV chose to yield. The AV accelerates with the intention of giving priority through the intersection within 0 to 1.5 seconds. Nevertheless, as the HV did not slow down to give way, the emergency braking was triggered at 2 seconds and continued until 3 seconds and the AV had to slow down to yield.



(a)



(b)

Fig. 10. Case 3: the AV eventually crosses the intersection before the HV. Firstly, the AV attempted to pass through the intersection later by reducing its speed to avoid the HV within the first two seconds. But then the HV succumbed by actively slowing down, resulting in the AV to reverse its policy. As a result, the AV increased speed to prioritize through the intersection after 2.5 seconds.

Moreover, the policy has certain human-like characteristic and is more compatible with human driving habits.

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VI. Conclusion

In this paper, we present a game-theoretic decision-making framework with conflict-aware Nash equilibrium selection for autonomous vehicles at uncontrolled intersections. The closed-loop NEs are calculated based on dynamic programming and iterative best response. For the existence of multi-NE scenarios, we propose the concept of CAB-game and design a NE selection mechanism based on it. Combining the NE calculation and NE selection with a safety check module, the obtained policy can be used for autonomous vehicles to make optimal decisions through feedback during dynamic interactions in real time. Finally, the simulation and driving simulator experiments demonstrate that the proposed DGTD policy can performs well in interaction with human-driven vehicles and different intelligent driving policies. The proposed policy can reduce the average cost of AVs and the emergency braking rate, i.e., it strikes a good balance between safety and efficiency. This framework has potential in integrating autonomous vehicles into real traffic environments.

In the future, we will extend the proposed decision-making framework to more complex traffic scenarios, such as unprotected left-turn and continuous traffic flows. Stochastic factors will be studied to enhance the robustness of the method. Different driver types described by different cost functions and parameters will be further studied.

References

[1] L. Wei, Z. Li, J. Gong, C. Gong, and J. Li, "Autonomous driving strategies at intersections: Scenarios, state-of-the-art, and future outlooks," in *Proc. IEEE Int. Intell. Transp. Syst. Conf. (ITSC)*, Sep. 2021, pp. 44–51.

[2] J. Nan, W. Deng, and B. Zheng, "Intention prediction and mixed strategy Nash equilibrium-based decision-making framework for autonomous driving in uncontrolled intersection," *IEEE Trans. Veh. Technol.*, vol. 71, no. 10, pp. 10316–10326, Oct. 2022.

[3] B. Pinkovich, E. Rivlin, and H. Rotstein, "Predictive driving in an unstructured scenario using the bundle adjustment algorithm," *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 1, pp. 342–352, Jan. 2021.

[4] W. Schwarting, J. Alonso-Mora, and D. Rus, "Planning and decision-making for autonomous vehicles," *Annu. Rev. Control Robot. Auton. Syst.*, vol. 1, no. 1, pp. 187–210, 2018.

[5] Q. Zhang, R. Langari, H. E. Tseng, D. Filev, S. Szwabowski, and S. Coskun, "A game theoretic model predictive controller with aggressiveness estimation for mandatory lane change," *IEEE Trans. Intell. Vehicles*, vol. 5, no. 1, pp. 75–89, Mar. 2020.

[6] R. Chen and Z. Yang, "A cooperative merging strategy for connected and automated vehicles based on game theory with transferable utility," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 19213–19223, Oct. 2022.

[7] M. Liu, Y. Wan, F. L. Lewis, S. Nageshrao, and D. Filev, "A three-level game-theoretic decision-making framework for autonomous vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 20298–20308, Nov. 2022.

[8] N. Li, D. W. Oyler, M. Zhang, Y. Yildiz, I. Kolmanovsky, and A. R. Girard, "Game theoretic modeling of driver and vehicle interactions for verification and validation of autonomous vehicle control systems," *IEEE Trans. Control Syst. Technol.*, vol. 26, no. 5, pp. 1782–1797, Sep. 2018.

[9] B. Zhang and J. Hofbauer, "Equilibrium selection via replicator dynamics in 2×2 coordination games," *Int. J. Game Theory*, vol. 44, no. 2, pp. 433–448, May 2015.

[10] D. M. Ramsey and K. Szajowski, "Selection of a correlated equilibrium in Markov stopping games," *Eur. J. Oper. Res.*, vol. 184, no. 1, pp. 185–206, Jan. 2008.

[11] Y. Chen, J. Zha, and J. Wang, "An autonomous T-intersection driving strategy considering oncoming vehicles based on connected vehicle technology," *IEEE/ASME Trans. Mechatronics*, vol. 24, no. 6, pp. 2779–2790, Dec. 2019.

[12] S. Fu and H. Fu, "Modeling and TOPSIS-GRA algorithm for autonomous driving decision-making under 5G-V2X infrastructure," *Comput., Mater. Continua*, vol. 75, no. 1, pp. 1051–1071, 2023.

[13] B. Tang, S. Khokhar, and R. Gupta, "Turn prediction at generalized intersections," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2015, pp. 1399–1404.

[14] X. Hu, S. E. Li, and D. Kum, "Modeling and control problems in sustainable transportation and power systems," *Math. Problems Eng.*, vol. 2016, pp. 1–3, Jul. 2016.

[15] B. R. Kiran et al., "Deep reinforcement learning for autonomous driving: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 4909–4926, Jun. 2022.

[16] D. Isele, R. Rahimi, A. Cosgun, K. Subramanian, and K. Fujimura, "Navigating occluded intersections with autonomous vehicles using deep reinforcement learning," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2018, pp. 2034–2039.

[17] J. Chen, D. Sun, and M. Zhao, "Human-like control for automated vehicles and avoiding 'Vehicle face-Off' in unprotected left turn scenarios," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 2, pp. 1609–1618, Feb. 2023.

[18] J. Chen, S. E. Li, and M. Tomizuka, "Interpretable end-to-end urban autonomous driving with latent deep reinforcement learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 5068–5078, Jun. 2022.

[19] K. Shu, R. V. Mehrizi, S. Li, M. Pirani, and A. Khajepour, "Human inspired autonomous intersection handling using game theory," *IEEE Trans. Intell. Transp. Syst.*, vol. 1, no. 2, pp. 1–12, May 2023.

[20] K. Ji, N. Li, M. Orsag, and K. Han, "Hierarchical and game-theoretic decision-making for connected and automated vehicles in overtaking scenarios," *Transp. Res. Part C, Emerg. Technol.*, vol. 150, May 2023, Art. no. 104109.

[21] C. Zhao, D. Chu, Z. Deng, and L. Lu, "Human-like decision making for autonomous driving with social skills," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 9, pp. 12269–12284, Sep. 2024.

[22] P. Hang, C. Lv, C. Huang, J. Cai, Z. Hu, and Y. Xing, "An integrated framework of decision making and motion planning for autonomous vehicles considering social behaviors," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 14458–14469, Dec. 2020.

[23] S. Karimi and A. Vahidi, "Receding horizon motion planning for automated lane change and merge using Monte Carlo tree search and Level-K game theory," in *Proc. Amer. Control Conf. (ACC)*, Jul. 2020, pp. 1223–1228.

[24] N. Li, D. Oyler, M. Zhang, Y. Yildiz, A. Girard, and I. Kolmanovsky, "Hierarchical reasoning game theory based approach for evaluation and testing of autonomous vehicle control systems," in *Proc. IEEE 55th Conf. Decis. Control (CDC)*, Dec. 2016, pp. 727–733.

[25] V. G. Lopez, F. L. Lewis, M. Liu, Y. Wan, S. Nageshrao, and D. Filev, "Game-theoretic lane-changing decision making and payoff learning for autonomous vehicles," *IEEE Trans. Veh. Technol.*, vol. 71, no. 4, pp. 3609–3620, Apr. 2022.

[26] P. Mathieu and J.-P. Delahaye, "Softening and hardening in the iterated Prisoner's dilemma," *IEEE Trans. Syst. Man, Cybern. Syst.*, vol. 53, no. 2, pp. 654–663, Feb. 2023.

[27] C. W. Fox, F. Camara, G. Markkula, R. A. Romano, R. Madigan, and N. Merat, "When should the chicken cross the road–game theory for autonomous Vehicle–Human interactions," in *Proc. 4th Int. Conf. Vehicle Technol. Intell. Transp. Syst.*, 2018, pp. 431–439.

[28] E. Haruvy and D. O. Stahl, "Deductive versus inductive equilibrium selection: Experimental results," *J. Econ. Behav. Org.*, vol. 53, no. 3, pp. 319–331, Mar. 2004.

[29] J. C. Harsanyi and R. Selten, *A General Theory of Equilibrium Selection in Games*, vol. 1. Cambridge, MA, USA: MIT Press, 1988.

[30] P. Frasckiewicz, "On signaling games with quantum chance move," *J. Phys. A, Math. Theor.*, vol. 48, no. 7, Feb. 2015, Art. no. 075303.

[31] S. Morris, R. Rob, and H. S. Shin, "P-dominance and belief potential," *Econometrica*, vol. 63, no. 1, p. 145, Jan. 1995.

[32] B. Zhang and J. Hofbauer, "Quantal response methods for equilibrium selection in 2×2 coordination games," *Games Econ. Behav.*, vol. 97, pp. 19–31, May 2016.

[33] A. Dreves, "An algorithm for equilibrium selection in generalized Nash equilibrium problems," *Comput. Optim. Appl.*, vol. 73, no. 3, pp. 821–837, Jul. 2019.

[34] J. Duffy and D. Fehr, "Equilibrium selection in similar repeated games: Experimental evidence on the role of precedents," *Experim. Econ.*, vol. 21, no. 3, pp. 573–600, Sep. 2018.

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[35] F. Camara, S. Cosar, N. Bellotto, N. Merat, and C. W. Fox, "Continuous game theory pedestrian modelling method for autonomous vehicles," in *Human Factors in Intelligent Vehicles*. Denmark, Europe: River Publishers, 2022, pp. 1–20.

[36] M. Elhenawy, A. A. Elbery, A. A. Hassan, and H. A. Rakha, "An intersection game-theory-based traffic control algorithm in a connected vehicle environment," in *Proc. IEEE 18th Int. Conf. Intell. Transp. Syst.*, Sep. 2015, pp. 343–347, doi: 10.1109/ITSC.2015.65.

[37] C. Xu, W. Zhao, L. Li, Q. Chen, D. Kuang, and J. Zhou, "A Nash Q-Learning based motion decision algorithm with considering interaction to traffic participants," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 12621–12634, Nov. 2020.

[38] H. Hu, K. Nakamura, K. Hsu, N. Ehrich Leonard, and J. Fernández Fisac, "Emergent coordination through game-induced nonlinear opinion dynamics," in *Proc. 62nd IEEE Conf. Decis. Control (CDC)*, Dec. 2023, pp. 8122–8129.

[39] L. Peters, D. Fridovich-Keil, C. J. Tomlin, and Z. N. Sunberg, "Inference-based strategy alignment for general-sum differential games," 2020, *arXiv:2002.04354*.

[40] M. Ammour, R. Orjuela, and M. Basset, "A MPC combined decision making and trajectory planning for autonomous vehicle collision avoidance," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 24805–24817, Dec. 2022.

[41] W. B. Knox, A. Allievi, H. Banzhaf, F. Schmitt, and P. Stone, "Reward (Mis)design for autonomous driving," *Artif. Intell.*, vol. 316, Mar. 2023, Art. no. 103829.

[42] K. Liu, N. Li, H. E. Tseng, I. Kolmanovsky, A. Girard, and D. Filev, "Cooperation-aware decision making for autonomous vehicles in merge scenarios," in *Proc. 60th IEEE Conf. Decis. Control (CDC)*, Dec. 2021, pp. 5006–5012.

[43] Z. Deng et al., "A probabilistic model for driving-style-recognition-enabled driver steering behaviors," *IEEE Trans. Syst. Man, Cybern., Syst.*, vol. 52, no. 3, pp. 1838–1851, Mar. 2022.

[44] W. Hu et al., "Formulating vehicle aggressiveness towards social cognitive autonomous driving," *IEEE Trans. Intell. Vehicles*, vol. 8, no. 3, pp. 2097–2109, Mar. 2023.

[45] D. Lacker, "On the convergence of closed-loop Nash equilibria to the mean field game limit," *Ann. Appl. Probab.*, vol. 30, no. 4, pp. 1693–1761, Aug. 2020.

[46] C. E. Lemke and J. T. Howson, "Equilibrium points of bimatrix games," *J. Soc. Ind. Appl. Math.*, vol. 12, no. 2, pp. 413–423, Jun. 1964.

[47] N. Li, I. Kolmanovsky, A. Girard, and Y. Yildiz, "Game theoretic modeling of vehicle interactions at unsignalized intersections and application to autonomous vehicle control," in *Proc. Annu. Amer. Control Conf. (ACC)*, Jun. 2018, pp. 3215–3220.

[48] R. Nagel, "Unraveling in guessing games: An experimental study," *Amer. Econ. Rev.*, vol. 85, no. 5, pp. 1313–1326, 1995.

[49] M. Yuan, J. Shan, and K. Mi, "Deep reinforcement learning based game-theoretic decision-making for autonomous vehicles," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 818–825, Apr. 2022.

[50] P. Hang, Y. Zhang, and C. Lv, "Brain-inspired modeling and decision-making for human-like autonomous driving in mixed traffic environment," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 10, pp. 1–13, May 2023.

[51] L. Cai, H. Guan, Q. H. Xu, X. Jia, and J. Zhan, "Game-theoretic decision-making method and motion planning for autonomous vehicles in overtaking," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 8, pp. 9693–9709, Aug. 2024.

[52] Y. Yan et al., "A multi-vehicle game-theoretic framework for decision making and planning of autonomous vehicles in mixed traffic," *IEEE Trans. Intell. Vehicles*, vol. 8, no. 11, pp. 4572–4587, Nov. 2023, doi: 10.1109/TIV.2023.3321346.

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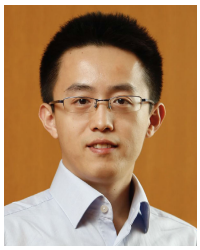

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